

Shared on-policy rollout

Both methods train on the same trajectory and the same original action tokens.

User task

"I want to return items I purchased."

Selected turn t : original action y_t

find_user_id_by_name_zip("John", "Doe", "12345")

Local observation o_t

Error: user not found

Later evidence

future turns and final outcome $R(\tau)$

HERO (ours)

Reflection compresses future evidence into a local hint

Reflector

sees the full trajectory τ and outcome $R(\tau)$
outputs a compact turn-level hint h_t

Teacher context

$H_t + R(\tau) + o_t + h_t$
compact hindsight, no raw future trajectory

✓ Targeted local credit

the hint summarizes only the relevant hindsight for this turn

Teacher signal on y_t

re-evaluates the same original action tokens with more actionable turn-level supervision

Vanilla full-demo teacher

Naive multi-turn extension of single-turn privileged distillation

Teacher context

$H_t + R(\tau) + \tau^*$
complete successful trajectory is visible at every turn

✗ Context mismatch

future actions and outcomes are unavailable to the student at early turns

Teacher signal on y_t

may rely on non-causal shortcuts, giving weak or misleading token-level supervision